

Figure 5: Choose the file system type that contains the deleted files

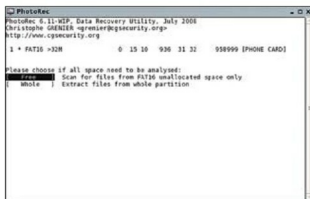


Figure 6: Recover files from the whole partition or from 'free space'

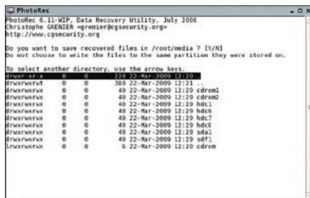


Figure 7: Choose the destination to save recovered files

file types that can be recovered. Use the up/down arrow keys to move between entries, and use the *Space* bar to select [x] or unselect [] file types to be recovered. For instance, to recover only picture files, choose 'jpg', 'gif', 'dsc', etc (Figure 4). Select 'Quit' when done. This takes you back to the last screen. Select the 'Search' option.

Step 5: To recover lost/deleted files, PhotoRec needs to know the file system type your files were stored in. Options include 'ext2/ext3' and 'Other'. Choose 'Other' for FAT, NTFS, ReiserFS, etc (Figure 5).

Step 6: The next screen gives the option of choosing between recovering data from the 'whole partition/disk' or from the 'free space' of the partition

(Figure 6). This 'free space' has the inodes [en.wikipedia.org/wiki/Inode] that contain the deleted data. Choose the 'free space' option using the up/down arrow keys.

Step 7: PhotoRec now needs to know the destination folder to save the recovered files. In Step 1, we created a folder called *photorec_files*. Navigate to your path using the up/down arrow key and press *Enter*. Mine was */hdc6/home/nelson/photorec_files*. If the path is not provided, the default directory will be */root*.

Your recovered files will be saved in the destination directory in a number of folders *recup_dir.1*, *recup_dir.2*, etc. In a single *recup_dir*, you will find zip files, doc files, jpg files, etc, if you have chosen to recover these files.

Searching for a file through those that are lost can be a real pain. You need to sort these files out. Here is what you can do to sort out zip files. Make a directory for zip files as follows:

```
mkdir /home/user/Zip
```

Now as the root user:

```
mv /home/user/recup_dir.1/*.*zip ~/Zip
```

Alternatively, issue the following command:

```
mv /home/user/recup_dir.1/*.*zip ~/Zip
```

You can similarly repeat the steps for other file types. Of particular interest is how to sort out picture files. Let us separate those little thumbnail pictures from your 'real' ones. Again, create a directory for small pictures with the code below:

```
mkdir ~/small_jpg
```

Now, as the root user, issue the following command:

```
find /home/user/recup_dir.1/ -name "*.jpg" \
-size -20K | xarg -i mv {} /home/user/small_jpg
```

This will find all jpg files equal to or smaller than 20K and move them to *small_jpg*.

The bottom line

I found almost all of my deleted photographs and various other files with a little sweat (manually opening each file till I got the one I was really looking for) using PhotoRec. Anyone looking at a launch pad for a career in forensics? 

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